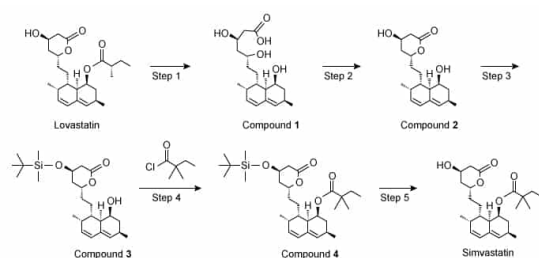


High blood cholesterol (hypercholesterolemia) accounts for as much as one third of global ischemic heart disease cases. Since all cholesterol has either a biosynthetic or dietary source, drugs that attenuate cholesterol biosynthesis are attractive options to treat patients with high blood cholesterol. Statins are an important class of drugs that competitively inhibit 3-hydroxy-3-methylglutaryl coenzyme A (HMG-CoA) reductase, the critical regulatory enzyme in the cholesterol biosynthesis pathway. All statins contain a structural component that mimics the structure of HMG-CoA, the six-carbon natural substrate of HMG-CoA reductase, and additional lipophilic regions that contour to the three-dimensional shape of the enzyme active site. The first statins that appeared on the market were bacterial natural products. Later, more potent statin derivatives were developed through semisynthetic techniques.

Lovastatin is a natural fungal product isolated from *Aspergillus terreus*, which is used as the starting material in the semisynthetic preparation of simvastatin, a derivative with improved HMG-CoA inhibitory activity relative to lovastatin. An overview of the five-step synthetic conversion of lovastatin to simvastatin is depicted in Scheme 1.



Scheme 1 Semisynthetic preparation of simvastatin from lovastatin

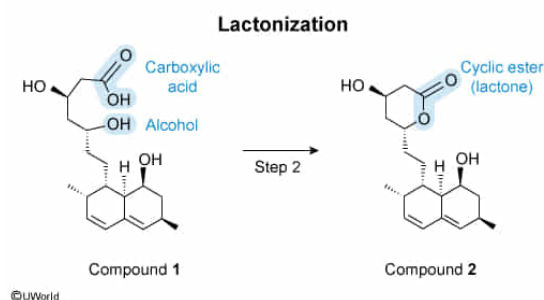
What type of chemical transformation is taking place in Step 2?

- ☐ A. Hydrolysis [19%]
- ✓ ☒ B. Lactonization [58%]
- ☐ C. Reduction [11%]
- ☐ D. Lactamization [9%]

Correct

58% answered correctly

Explanation:



Lactonization is the intramolecular reaction between an alcohol and a carboxylic acid that creates a **cyclic ester (lactone)**. Although some lactonization reactions can be spontaneous,

most require the addition of either an **acidic catalyst** or **agent**, which provides the carboxylic acid with a better leaving group. Lactonization can be classified as both a dehydration reaction and a **condensation reaction**, as a molecule of water is lost when the two functional groups combine to form a lactone. Lactonizations that result in the formation of five- or six-membered rings are especially favorable.

The reaction depicted in Step 2 of Scheme 1 shows an alcohol and a carboxylic acid in the reactant (Compound **1**), and a cyclic ester in the product (Compound **2**). This reaction is an example of a lactonization.

(Choice A) A hydrolysis reaction is defined as a reaction in which a molecule of water is consumed to cleave a bond. The reaction in Step 2 is the opposite of this process.

(Choice C) In organic chemistry, a **reduction** reaction occurs when a substance gains electrons or decreases the number of carbon-heteroatom bonds at a specific carbon atom. The reaction in Step 2 does not change the number of carbon-heteroatom bonds for the carboxylic acid carbon or the carbon bearing the alcohol.

(Choice D) A lactamization is an intramolecular condensation reaction between an amine and a carboxylic acid that forms a **cyclic amide (lactam)**. The reaction in Step 2 depicts a reaction between an *alcohol* and a carboxylic acid.

Educational objective:

Organic transformations are classified by the types of functional groups that react and form. A lactonization is a condensation reaction between an alcohol and a carboxylic acid that forms a cyclic ester (lactone).

Time spent:0 sec

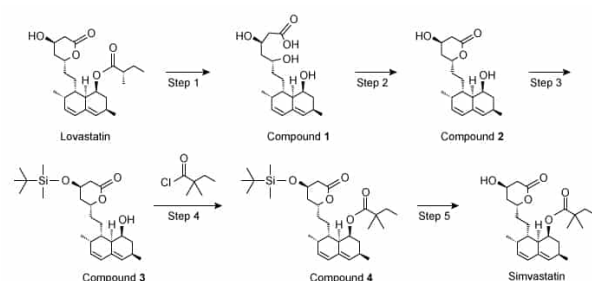
Subject:
Organic Chemistry

QID:403790

System:
Functional Groups and Their Reactions

High blood cholesterol (hypercholesterolemia) accounts for as much as one third of global ischemic heart disease cases. Since all cholesterol has either a biosynthetic or dietary source, drugs that attenuate cholesterol biosynthesis are attractive options to treat patients with high blood cholesterol. Statins are an important class of drugs that competitively inhibit 3-hydroxy-3-methylglutaryl coenzyme A (HMG-CoA) reductase, the critical regulatory enzyme in the cholesterol biosynthesis pathway. All statins contain a structural component that mimics the structure of HMG-CoA, the six-carbon natural substrate of HMG-CoA reductase, and additional lipophilic regions that contour to the three-dimensional shape of the enzyme active site. The first statins that appeared on the market were bacterial natural products. Later, more potent statin derivatives were developed through semisynthetic techniques.

Lovastatin is a natural fungal product isolated from *Aspergillus terreus*, which is used as the starting material in the semisynthetic preparation of simvastatin, a derivative with improved HMG-CoA inhibitory activity relative to lovastatin. An overview of the five-step synthetic conversion of lovastatin to simvastatin is depicted in Scheme 1.



Scheme 1 Semisynthetic preparation of simvastatin from lovastatin

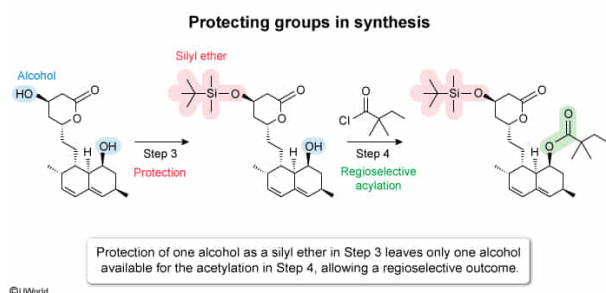
The transformation taking place in Step 3 is necessary to:

- ☐ A. deprotect the alcohol and ensure the stereoselectivity of the Step 4 acylation. [2%]
- ☐ B. protect the alcohol and ensure the stereoselectivity of the Step 4 acylation. [19%]
- ☐ C. deprotect the alcohol and ensure the regioselectivity of the Step 4 acylation. [4%]
- ✔ ☒ D. protect the alcohol and ensure the regioselectivity of the Step 4 acylation. [73%]

Correct

73% answered correctly

Explanation:



Protecting groups are functional groups that can be reversibly added to a substrate to ensure the integrity of a later (target) synthetic step. Some organic transformations are not readily accomplished in the presence of other functional groups or multiple copies of a single functional group. In these cases, the problematic functional groups are first converted to an appropriate protecting group that will not react under the conditions of the target reaction. After the target reaction has been achieved, **protecting groups can be removed (deprotection)** to restore the original functional group structure.

The nucleophilicity of alcohols frequently requires that they be protected during multistep syntheses. **Silyl ethers** are one of the most common classes of alcohol protecting groups, and the *tert*-butyl-dimethylsilyl (TBDMS) variant has become one of the most utilized alcohol protecting groups. TBDMS groups can be safely introduced using several reagents and are easily removed using chemistry that does not impact most other functional groups.

Organic transformations can demonstrate several types of selectivity. A **stereoselective reaction** results in the preferential formation of a stereoisomer. For example, a reaction that prefers either a *cis* or *trans* outcome would be stereoselective. A regioselective reaction results in the preferential reaction at one location within a molecule, often resulting in constitutional isomers.

In Step 3, the alcohol in Compound **2** is protected as a silyl ether (Compound **3**). Without this step, only two possible alcohol groups would react with the acid chloride in Step 4. By **protecting** one of the alcohols as a silyl ether, the other alcohol could be **regioselectively acylated** in Step 4.

(Choices A and B) None of the steps in the synthesis result in the formation of new stereocenters. Consequently, stereoselectivity does not factor into the use of protecting groups in this question.

(Choice C) Step 3 depicts the *addition* of a functional group to the alcohol, rather than a removal. Consequently, Step 3 is *not* a deprotection.

Educational objective:

Discrete steps in a multistep synthesis are generally included to achieve a specific

synthetic purpose. Protecting groups can be added or removed to promote enhanced selectivity (eg, stereoselectivity, regioselectivity) for later synthetic steps.

Time spent:0 sec

Subject:

Organic Chemistry

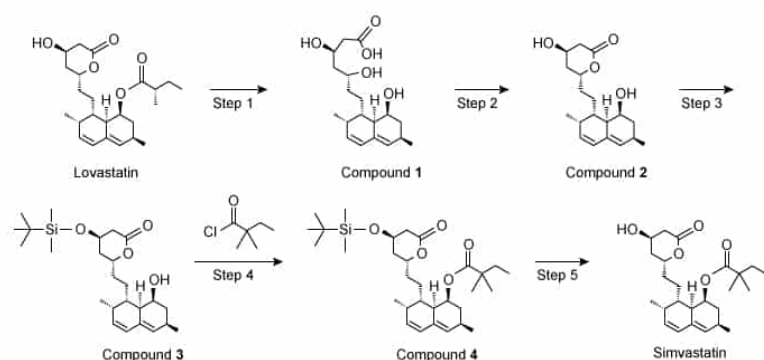
QID:403791

System:

Functional Groups and Their Reactions

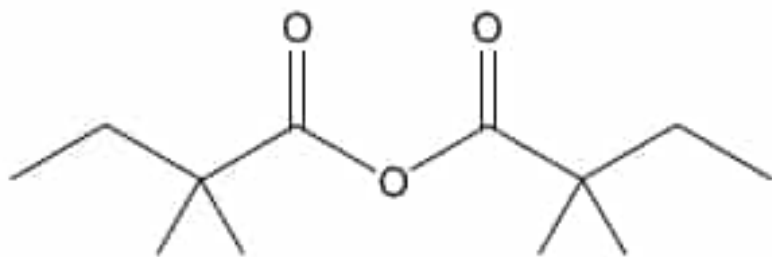
High blood cholesterol (hypercholesterolemia) accounts for as much as one third of global ischemic heart disease cases. Since all cholesterol has either a biosynthetic or dietary source, drugs that attenuate cholesterol biosynthesis are attractive options to treat patients with high blood cholesterol. Statins are an important class of drugs that competitively inhibit 3-hydroxy-3-methylglutaryl coenzyme A (HMG-CoA) reductase, the critical regulatory enzyme in the cholesterol biosynthesis pathway. All statins contain a structural component that mimics the structure of HMG-CoA, the six-carbon natural substrate of HMG-CoA reductase, and additional lipophilic regions that contour to the three-dimensional shape of the enzyme active site. The first statins that appeared on the market were bacterial natural products. Later, more potent statin derivatives were developed through semisynthetic techniques.

Lovastatin is a natural fungal product isolated from *Aspergillus terreus*, which is used as the starting material in the semisynthetic preparation of simvastatin, a derivative with improved HMG-CoA inhibitory activity relative to lovastatin. An overview of the five-step synthetic conversion of lovastatin to simvastatin is depicted in Scheme 1.



Scheme 1 Semisynthetic preparation of simvastatin from lovastatin

What is the IUPAC name for the following molecule, which could be used as an alternative reagent for the acylation in Step 4?



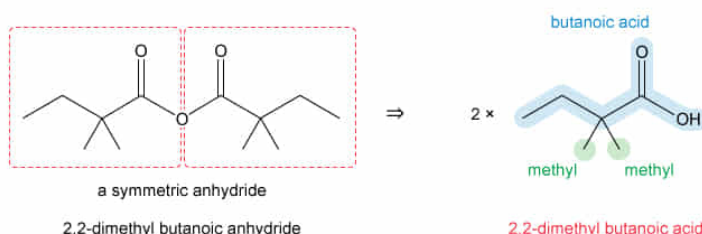
- ✓ ☒ A. 2,2-dimethylbutanoic anhydride [83%]
☐ B. 2-ethyl-2-methylpropanoic anhydride [5%]
☐ C. 2,2-dimethylbutanoic ether [4%]
☐ D. 2,2-dimethylbutanoic ester [6%]

Correct

83% answered correctly

Explanation:

Carboxylic acid anhydride IUPAC nomenclature



©UWorld

An **anhydride** is a functional group that results from the **condensation of two carboxylic acids**. Functionally, the formation of an anhydride from carboxylic acids is a dehydration reaction that results in the loss of water. However, anhydrides are often synthesized by the **reaction** of a carboxylic acid with an acid halide, an activated form of the second carboxylic acid.

Anhydride **IUPAC nomenclature** guidelines are based on the

identity of the **source carboxylic acids**, followed by the word "**anhydride**." **Symmetric anhydrides** are made from two identical carboxylic acids and are named by stating the source carboxylic acid (with an *-oic* suffix) followed by the word "anhydride." Asymmetric (mixed) anhydrides are made from two different carboxylic acids and are named by stating the source carboxylic acids in alphabetical order (each with an *-oic* suffix) followed by the word "anhydride."

In this question, the two halves of the anhydride functional group are identical, and the molecule is a symmetric anhydride. The longest continuous carbon chain in the source carboxylic acid contains four carbons (butanoic acid). Each butanoic acid source molecule contains two methyl substituents on the carbon adjacent to the carbonyl. Therefore, the full name of the source carboxylic acid is 2,2-dimethylbutanoic acid. The IUPAC name for the symmetric anhydride replaces the word "acid" with "anhydride" to give **2,2-dimethylbutanoic anhydride**.

(Choice B) In 2-ethyl-2-methylpropanoic anhydride, the source acid (parent chain) is incorrectly identified as a three-carbon chain (propanoic acid) with a 2-ethyl group.

(Choice C) Although an **ether** joins two carbon chains by an oxygen atom like an anhydride, an ether lacks the carbonyl groups adjacent to the central oxygen atom found in an anhydride.

(Choice D) An **ester** is the condensation product of a carboxylic acid with an alcohol. Although an ester shares some structural characteristics with an anhydride, an ester lacks the presence of a second carbonyl group adjacent to the central oxygen found in an anhydride.

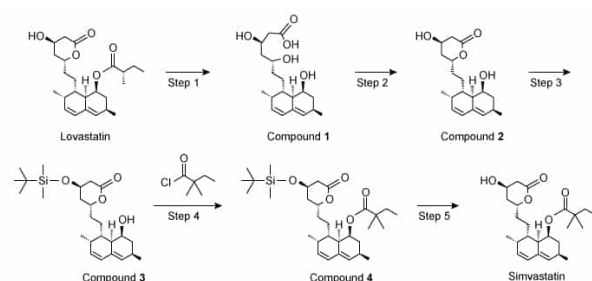
Educational objective:

Anhydrides are formed from a condensation reaction between two carboxylic acid molecules. IUPAC names for anhydrides include the source carboxylic acids followed by the word "anhydride."

Time spent:0 sec**QID:**403792**Subject:**
Organic Chemistry**System:**
Functional Groups and Their Reactions

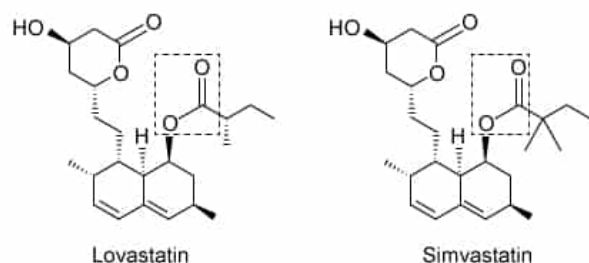
High blood cholesterol (hypercholesterolemia) accounts for as much as one third of global ischemic heart disease cases. Since all cholesterol has either a biosynthetic or dietary source, drugs that attenuate cholesterol biosynthesis are attractive options to treat patients with high blood cholesterol. Statins are an important class of drugs that competitively inhibit 3-hydroxy-3-methylglutaryl coenzyme A (HMG-CoA) reductase, the critical regulatory enzyme in the cholesterol biosynthesis pathway. All statins contain a structural component that mimics the structure of HMG-CoA, the six-carbon natural substrate of HMG-CoA reductase, and additional lipophilic regions that contour to the three-dimensional shape of the enzyme active site. The first statins that appeared on the market were bacterial natural products. Later, more potent statin derivatives were developed through semisynthetic techniques.

Lovastatin is a natural fungal product isolated from *Aspergillus terreus*, which is used as the starting material in the semisynthetic preparation of simvastatin, a derivative with improved HMG-CoA inhibitory activity relative to lovastatin. An overview of the five-step synthetic conversion of lovastatin to simvastatin is depicted in Scheme 1.



Scheme 1 Semisynthetic preparation of simvastatin from lovastatin

The structures of lovastatin and simvastatin are shown. Which molecule will have a higher rate of basic hydrolysis for the indicated ester?

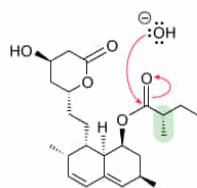


- ✔ ☒ A. Lovastatin [78%]
☐ B. Simvastatin [12%]
☐ C. Both compounds have equivalent rates of basic ester hydrolysis. [8%]
☐ D. Nothing can be inferred about the rate of basic ester hydrolysis. [1%]

Explanation:

Impact of α -methyl groups on nucleophilic attack

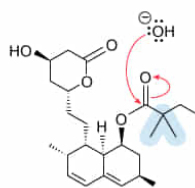
One alpha methyl group makes the ester **less sterically hindered** to nucleophilic attack



Lovastatin

Faster rate of hydrolysis

Two alpha methyl groups make the ester **more sterically hindered** to nucleophilic attack



Simvastatin

Slower rate of hydrolysis

An **ester** is a **carboxylic acid derivative** formed through the reaction of an alcohol with a carboxylic acid (esterification). An **esterification** is a condensation reaction that results in the loss of a molecule of water. Consequently, **ester hydrolysis** is the reciprocal chemical process to esterification in which a molecule of water is consumed to cleave the carbon-oxygen single bond in an ester. Ester hydrolysis follows an addition-elimination mechanism and typically requires the presence of either a basic catalyst or an acidic catalyst.

In acid-catalyzed ester hydrolysis, a catalytic amount of strong acid protonates the oxygen of the ester carbonyl and makes the carbonyl carbon a better electrophile for the nucleophilic addition of water. In the elimination half of the mechanism, the acid catalyst protonates the alkoxy group to make it a better **leaving group**. In a **base-catalyzed ester hydrolysis**, a catalytic amount of strong base directly adds to the electrophilic carbon of the ester carbonyl. Elimination of an alkoxide, followed by proton transfers, generates alcohol and carboxylate hydrolysis products.

The structures of lovastatin and simvastatin differ only by one methyl group at the alpha-position relative to the ester carbonyl. The second α -methyl group in simvastatin provides additional **steric hindrance**, which limits the accessibility of nucleophilic attack at the adjacent carbonyl carbon. In contrast, the structure of lovastatin contains only one α -methyl group to the ester carbonyl and has greater susceptibility to nucleophilic attack. Consequently, the **rate of ester hydrolysis** will be **higher** for the **less-substituted molecule, lovastatin**.

(Choice B) Simvastatin contains an additional α -methyl group relative to lovastatin, which provides additional steric hindrance to nucleophilic attack. Consequently, the rate of ester hydrolysis will be *lower* for simvastatin.

(Choices C and D) The rate of carboxylic acid hydrolysis is impacted by molecular structure and the susceptibility to nucleophilic attack. Lovastatin and simvastatin differ in the number of α -methyl groups adjacent to the carbonyl and are expected to have *different* rates of ester hydrolysis.

Educational objective:

The rate of ester hydrolysis is impacted by molecular structure, particularly at alpha-positions to the ester carbonyl. Esters with greater steric hindrance at the alpha-position have reduced susceptibility to nucleophilic attack and have slower rates of hydrolysis.

Time spent:0 sec**Subject:**

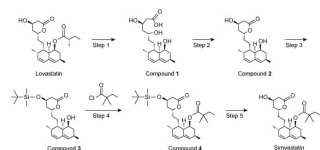
Organic Chemistry

QID:403793**System:**

Functional Groups and Their Reactions

High blood cholesterol (hypercholesterolemia) accounts for as much as one third of global ischemic heart disease cases. Since all cholesterol has either a biosynthetic or dietary source, drugs that attenuate cholesterol biosynthesis are attractive options to treat patients with high blood cholesterol. Statins are an important class of drugs that competitively inhibit 3-hydroxy-3-methylglutaryl coenzyme A (HMG-CoA) reductase, the critical regulatory enzyme in the cholesterol biosynthesis pathway. All statins contain a structural component that mimics the structure of HMG-CoA, the six-carbon natural substrate of HMG-CoA reductase, and additional lipophilic regions that contour to the three-dimensional shape of the enzyme active site. The first statins that appeared on the market were bacterial natural products. Later, more potent statin derivatives were developed through semisynthetic techniques.

Lovastatin is a natural fungal product isolated from *Aspergillus terreus*, which is used as the starting material in the semisynthetic preparation of simvastatin, a derivative with improved HMG-CoA inhibitory activity relative to lovastatin. An overview of the five-step synthetic conversion of lovastatin to simvastatin is depicted in Scheme 1.



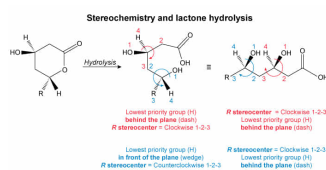
Scheme 1 Semisynthetic preparation of simvastatin from lovastatin

If the lactone in lovastatin is hydrolyzed, what will be the structure of the resultant hydrolysis product?

- ☒ A.
- ☐ B.
- ☐ C.
- ☐ D.

Correct
34% answered correctly

Explanation:



Like the [hydrolysis of an ester](#), lactone hydrolysis is a chemical transformation in which a molecule of water breaks the carbon-oxygen single bond in an ester to yield an alcohol and a carboxylic acid. Since a lactone is a cyclic functional group, lactone hydrolysis results in a single product that contains both the alcohol and carboxylic acid functional groups. Although lactone hydrolysis still requires an acidic or basic catalyst, catalytic conditions can often be milder given that lactone hydrolysis can relieve ring strain.

According to Cahn-Ingold-Prelog guidelines, the absolute configuration of a chiral carbon atom is determined by the relative spatial arrangement of groups attached to the chiral carbon. Groups are prioritized using a [systematic process](#) based on atomic number that accounts for molecular connectivity, the possibility of tiebreakers, and the presence of multiple bonds. Group priorities are used to assign each chiral carbon atom either an *R* or *S* configuration.

The initial product of hydrolysis is depicted with the same orientation as the initial lactone reactant. When Cahn-Ingold-Prelog priority guidelines are applied, the hydrolysis product is shown to have two chiral carbon atoms, each with an *R* configuration. Transposing the Cahn-Ingold-Prelog priority assignments and *R* configuration to the scaffold drawn in the same perspective as the answer choices shows that the hydroxyl group formed by lactone hydrolysis should be oriented out of the plane (ie, wedge). The structure shown in Choice A correctly depicts the stereochemistry of the lactone hydrolysis product.

(Choices B and C) One of the chiral carbons has incorrect stereochemical orientation in each of these structures.

(Choice D) The stereochemical orientation is incorrect for both chiral carbons.

Educational objective:

Lactone hydrolysis generates a single molecule containing both alcohol and carboxylic acid functional groups. The stereochemistry of all other carbon atoms is retained in hydrolysis products.

Time spent: 0 sec
Subject:
Organic Chemistry

QID:403794
System:
Functional Groups and Their Reactions